

WT01P4C5-S1

Datasheet



Version 1.3

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V1.1	2025-9-25	Correct one error in 2.2 Hardware Parameters	Pail	Louie
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1. Overview

1.1. Products Introduction

WT01P4C5-S1, a core board with integrated 2.4GHz & 5GHz Wi-Fi6 and NOR FLASH based on Espressif's ESP32-P4 and ESP32-C5 series chips designed by Wireless-Tag Technology Co., Limited. The core processor chip, ESP32-P4, is stackable with either 16MB or 32MB PSRAM in a package that includes a high-performance (HP) system and a low-power (LP) system. The HP system is a RISC-V dual-core processor running at 360MHz and includes a JPEG codec, Pixel-Processing Accelerator, H.264 video encoder, and MIPI interfaces, providing powerful image and voice processing capabilities. The ESP32-C5 high-performance SOC on the core board supports 2.4 & 5G dual-band Wi-Fi6 (802.11ax), Bluetooth®5 (LE), Zigbee, and Thread (802.15.4). The ESP32-P4 and ESP32-C5 on the core board are connected via their SDIO interface, and the rest of the pins are already pinned out.

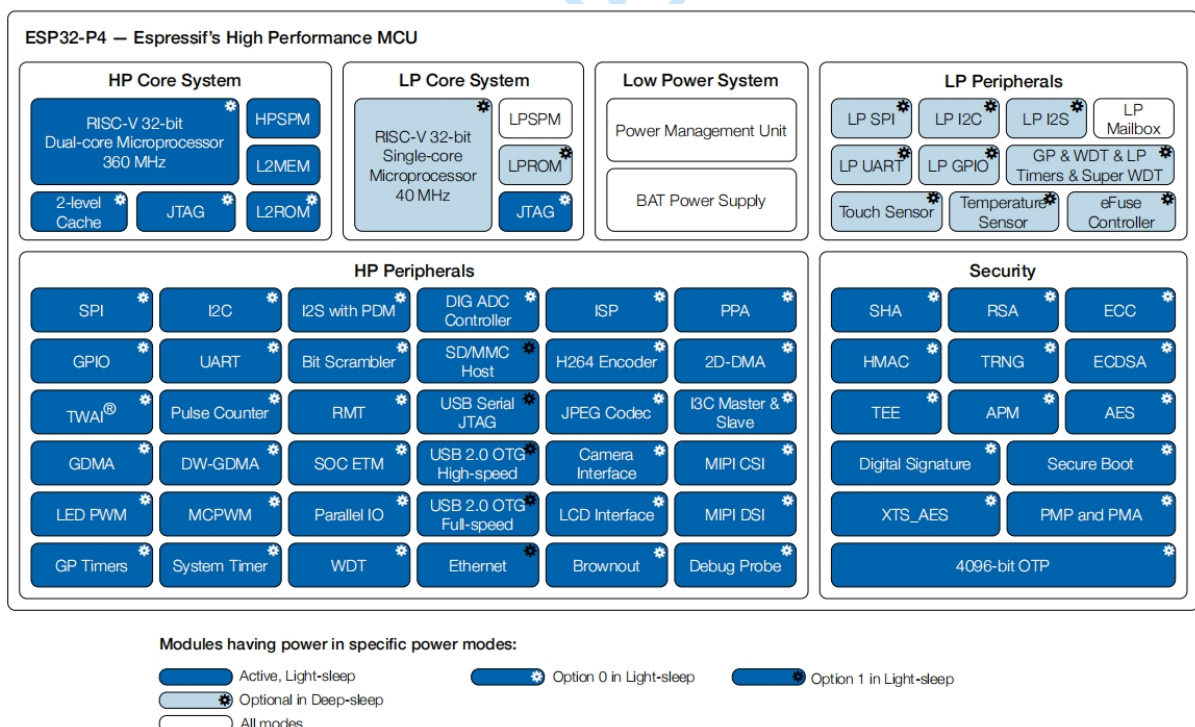


Figure 1: Main Chip Architecture Diagram

The WT01P4C5-S1 series is available in two sizes, see the table below for more information.

WT01P4C5-S1 Series Model Number Comparison

Part Number	Flash	Psram	Module Size (mm)
WT01P4C5-S1-N16R16	16MB	16MB	35.00*35.00
WT01P4C5-S1-N16R32	16MB	32MB	35.00*35.00

1.2.Product Features

- Dual-core 360 MHz high-frequency CPU
- 16 MB Flash and 16/32 MB Psram
- supports 2.4GHz & 5GHz dual-band Wi-Fi6, BLE5.3, Zigbee, Thread
- ESP32-P4 and ESP32-C5 chip with full pinout
- Supports multiple multimedia interfaces
- Core board size is small, easy to hardware design
- Development materials are complete

1.3.Product Pictures



Figure 2:WT01P4C5-S1(front)



Figure 3:WT01P4C5-S1 (back)



Figure 4:WT01P4C5-S1(front)



Figure 5:WT01P4C5-S1(back)

1.4.Application Scenarios

- Smart Home
- Industrial Automation
- Consumer Electronics

- HMI Human Machine Interaction
- Electronic Robotics
- Camera Video Streaming
- USB Devices
- Healthcare

2. Product Specification

2.1. Block Diagram

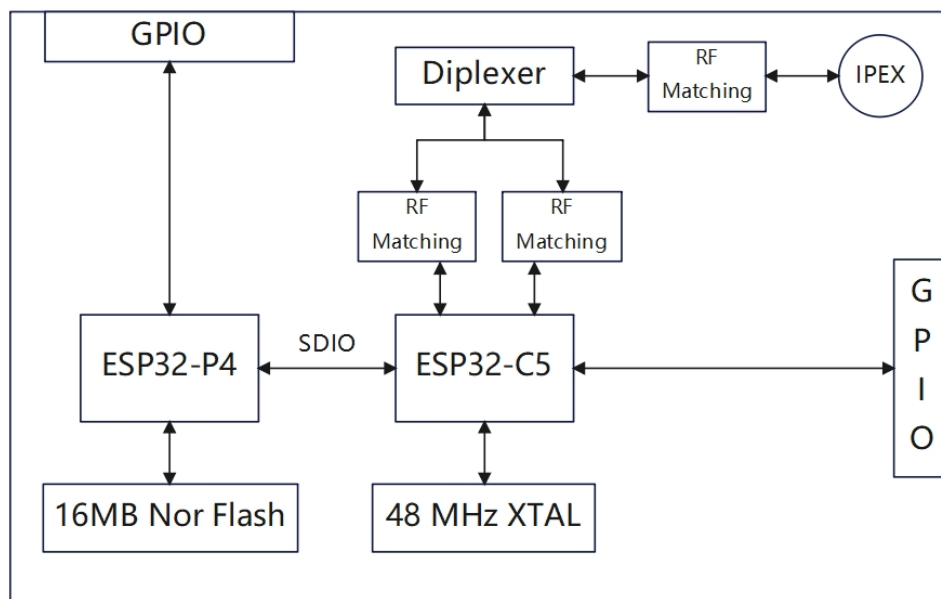


Figure 6: Block diagram of WT01P4C5-S1

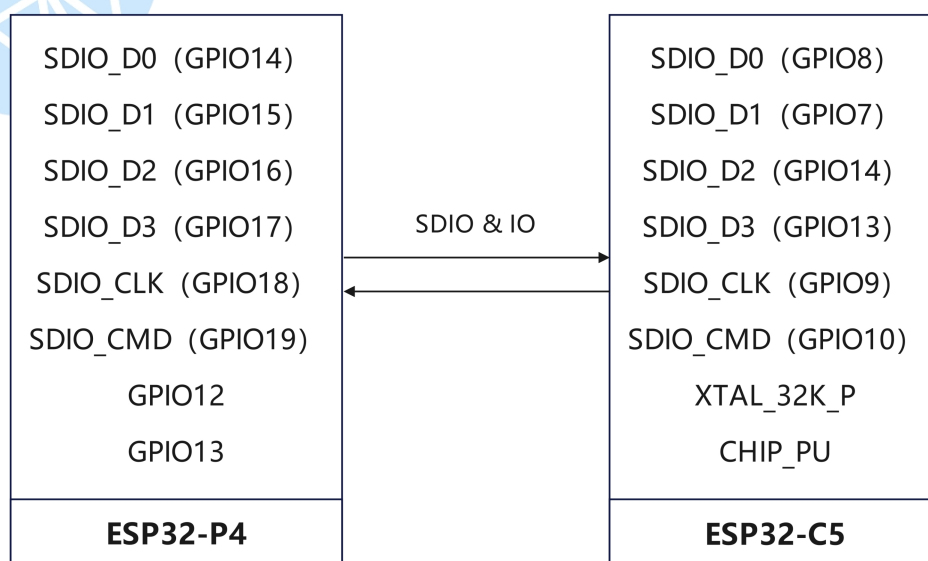


Figure 7: P4 & C5 Pin Correspondence Diagram

2.2. Hardware Parameters

ESP32-P4	Core	32-bit RISC-V dual-core
	Main Frequency	360 MHz (HP Core)
		40 MHz (LP Core)
ESP32-C5	Core	32-bit RISC-V dual-core
	Main Frequency	240 MHz
Memory	ESP32-P4 ROM	128 KB HP ROM
		16 KB LP ROM
	ESP32-P4 SRAM	768 KB HP L2MEM
		32 KB LP SRAM
	ESP32-P4 PSRAM	16/32 MB
	ESP32-C5 ROM	320 KB
	ESP32-C5 SRAM	384 KB
	Flash	16MB
ESP32-P4 Peripheral Interface	GPIO	46
	SPI	2
	LP SPI	1
	UART	5
	LP UART	1
	I3C	1
	I2C	2
	LP I2C	1
	I2S	3
	LP I2S	1
	USB JTAG	1
	LED PWM	1
	MCPWM	2

	TWAI®Controller (compatible with ISO 11898-1)	3
	Hight-Speed USB 2.0 OTG	1
	Full-Speed USB 2.0 OTG	1
	100 Mbit Ethernet	1
	MIPI CSI-2	1
	MIPI DSI	1
	Parallel IO interface	1
	12-bit multi-channel ADC	2
	Temperature sensor	1
	Touch sensor	1
	Analog voltage comparator	1
	Brown-out detector	1
ESP32-C5 Peripheral Interface	GPIO	12
	SPI	1
	UART	2
	I2C	1
	I2S	1
	LED PWM	1
	12-bit multi-channel ADC	1
	Temperature sensor	1
Image and Voice Processing Functionality	JPEG Codec	1
	PPA	1
	ISP	1
	H264 encoder	1

3. Pin Definitions

3.1.Pin Layout

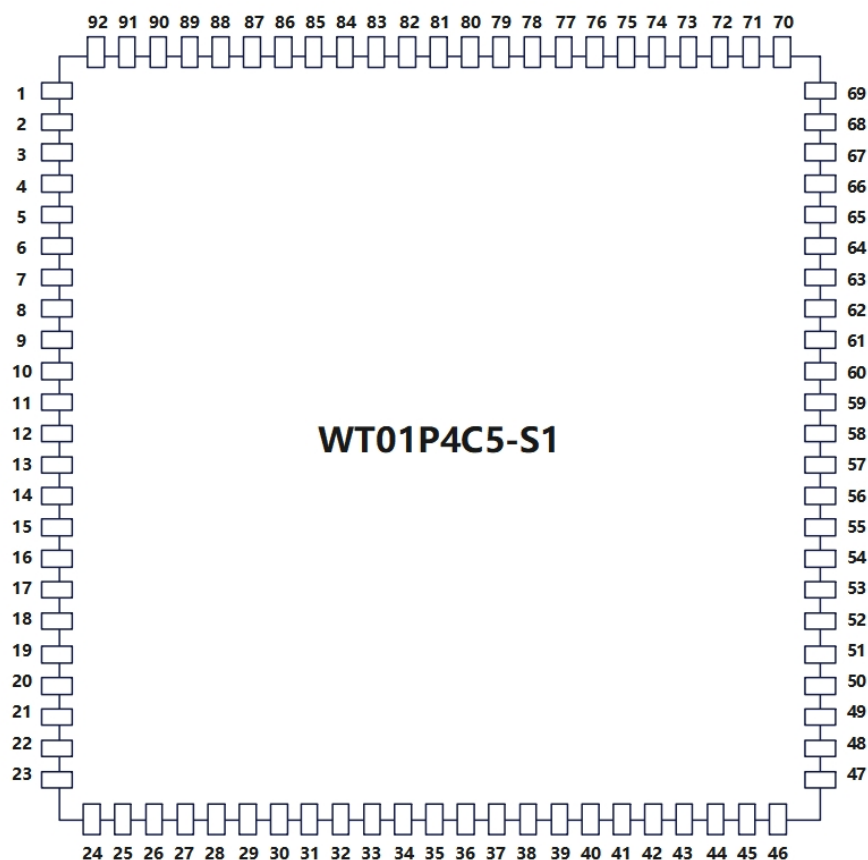


Figure 7:WT01P4C5-S1 Pin Layout

3.2.Pin Description

Pin Definitions

No.	Name	Function
1	C5_EN	Enable ESP32-C5 chip (internal 10K pull-up)
2	C5_IO1	GPIO1, XTAL_32K_N, LP_GPIO1, LP_UART_DSRN, ADC1_CHO
3	C5_IO2	GPIO2, MTMS, LP_GPIO2, LP_UART_RTSN, ADC1_CH1, FSPIQ
4	C5_IO3	GPIO3, MTDI, LP_GPIO3, LP_UART_CTSN, ADC1_CH2
5	C5_IO4	LP_UART_RXD, LP_GPIO4, GPIO4, MTCK, ADC1_CH3, FSPIHD
6	C5_IO5	LP_UART_TXD, LP_GPIO5, GPIO5, MTDO, ADC1_CH4, FSPIWP
7	C5_IO6	LP_I2C_SDA, LP_GPIO6, GPIO6, ADC1_CH5, FSPICLK
8	C5_U0TXD	GPIO11, UOTXD(ESP32-C5 Download Pin)
9	C5_U0RXD	GPIO12, UORXD(ESP32-C5 Download Pin)

10	GND	GROUND
11	P4_EN	Enable ESP32-P4 chip (internal 10K pull-up)
12	GPIO0	GPIO0, LP_GPIO0, XTAL_32K_N
13	GPIO1	GPIO1, LP_GPIO1, XTAL_32K_P
14	GPIO2	GPIO2, MTCK, LP_GPIO2, TOUCH_CHANNEL0
15	GPIO3	GPIO3, MTDI, LP_GPIO3, TOUCH_CHANNEL1
16	GPIO4	GPIO4, MTMS, LP_GPIO4, TOUCH_CHANNEL2
17	NC1	No internal connections, physical presence only
18	GPIO6	GPIO6, SPI2_HOLD_PAD, LP_GPIO6, TOUCH_CHANNEL4
19	GPIO7	GPIO7, SPI2_CS_PAD, LP_GPIO7, TOUCH_CHANNEL5
20	GPIO8	GPIO8, UART0_RTS_PAD, SPI2_D_PAD, LP_GPIO8, TOUCH_CHANNEL6
21	GPIO9	GPIO9, UART0_CTS_PAD, SPI2_CK_PAD, LP_GPIO9, TOUCH_CHANNEL7
22	GPIO10	GPIO10, UART1_TXD_PAD, SPI2_Q_PAD, LP_GPIO10, TOUCH_CHANNEL8
23	NC2	No internal connections, physical presence only
24	GPIO11	GPIO11, UART1_RXD_PAD, SPI2_WP_PAD, LP_GPIO11, TOUCH_CHANNEL9
25	GPIO20	GPIO20, ADC1_CHANNEL4
26	GPIO21	GPIO21, ADC1_CHANNEL5
27	GPIO22	GPIO22, ADC1_CHANNEL6
28	GPIO23	GPIO23, ADC1_CHANNEL7, REF_50M_CLK_PAD
29	GND	GROUND
30	DSI_DATAP1	MIPI DSI PHY DATAP1
31	DSI_DATAN1	MIPI DSI PHY DATAN1
32	DSI_CLKN	MIPI DSI PHY CLKN
33	DSI_CLKP	MIPI DSI PHY CLKP



34	DSI_DATAP0	MIPI DSI PHY DATAP0
35	DSI_DATAN0	MIPI DSI PHY DATAN0
36	GND	GROUND
37	CSI_DATAN0	MIPI CSI PHY DATAN0
38	CSI_DATAP0	MIPI CSI PHY DATAP0
39	CSI_CLKP	MIPI CSI PHY CLKP
40	CSI_CLKN	MIPI CSI PHY CLKN
41	CSI_DATAN1	MIPI CSI PHY DATAN1
42	CSI_DATAP1	MIPI CSI PHY DATAP1
43	GND	GROUND
44	USB_DM	USB2 OTG PHY DM
45	USB_DP	USB2 OTG PHY DP
46	GND	GROUND
47	GND	GROUND
48	GPIO24	GPIO24, USB1P1_N0
49	GPIO25	GPIO25, USB1P1_P0
50	GPIO26	GPIO26, USB1P1_N1
51	GPIO27	GPIO27, USB1P1_P1
52	GPIO28	GPIO28, SPI2_CS_PAD, GMAC_PHY_RXDV_PAD
53	GPIO29	GPIO29, SPI2_D_PAD, GMAC_PHY_RXD0_PAD
54	GPIO30	GPIO30, SPI2_CK_PAD, GMAC_PHY_RXD1_PAD
55	GPIO31	GPIO31, SPI2_Q_PAD, GMAC_PHY_RXER_PAD
56	GPIO32	GPIO32, SPI2_HOLD_PAD, GMAC_RMII_CLK_PAD
57	GPIO33	GPIO33, SPI2_WP_PAD, GMAC_PHY_TXEN_PAD
58	GPIO34	GPIO34, SPI2_IO4_PAD, GMAC_PHY_TXD0_PAD
59	GPIO35	GPIO35, SPI2_IO5_PAD, GMAC_PHY_TXD1_PAD(internal 10K pull-up)
60	GPIO36	GPIO36, SPI2_IO6_PAD, GMAC_PHY_TXER_PAD(internal 10K pull-up)
61	GPIO37	GPIO37, UART0_TXD_PAD, SPI2_IO7_PAD(ESP32-P4 Download Pin)

62	GPIO38	GPIO38, UART0_RXD_PAD, SPI2_DQS_PAD(ESP32-P4 Download Pin)
63	ESP_LDO_VO4	Output POWER (Output voltage range 0.5~2.7V or 3.3V, maximum output current 0.2A)
64	GPIO39	GPIO39, SD1_CDATA0_PAD, REF_50M_CLK_PAD
65	GPIO40	GPIO40, SD1_CDATA1_PAD, GMAC_PHY_TXEN_PAD
66	GPIO41	GPIO41, SD1_CDATA2_PAD, GMAC_PHY_TXD0_PAD
67	PWR_CTRL	Core board power control pin(default high, pull low to power off)
68	VCC_5V0	POWER (5V input for core board power supply)
69	VCC_5V0	POWER (5V input for core board power supply)
70	GND	GROUND
71	GND	GROUND
72	VBAT	battery power supply pin (reserved) No internal connections, physical presence only
73	GPIO42	GPIO42, SD1_CDATA3_PAD, GMAC_PHY_TXD1_PAD
74	GPIO43	GPIO43, SD1_CCLK_PAD, GMAC_PHY_TXER_PAD
75	GPIO44	GPIO44, SD1_CCMD_PAD, GMAC_RMII_CLK_PAD
76	GPIO45	GPIO45, SD1_CDATA4_PAD, GMAC_PHY_RXDV_PAD
77	GPIO46	GPIO46, SD1_CDATA5_PAD, GMAC_PHY_RXD0_PAD
78	GPIO47	GPIO47, SD1_CDATA6_PAD, GMAC_PHY_RXD1_PAD
79	GPIO48	GPIO48, SD1_CDATA7_PAD, GMAC_PHY_RXER_PAD
80	GPIO49	GPIO49, GMAC_PHY_TXEN_PAD, ADC2_CHANNEL0
81	GPIO50	GPIO50, GMAC_RMII_CLK_PAD, ADC2_CHANNEL1
82	GPIO51	GPIO51, GMAC_PHY_RXDV_PAD, ADC2_CHANNEL2, ANA_COMP0
83	GPIO52	GPIO52, GMAC_PHY_RXD0_PAD, ADC2_CHANNEL3, ANA_COMP0
84	GPIO53	GPIO53, GMAC_PHY_RXD1_PAD, ADC2_CHANNEL4, ANA_COMP1
85	GPIO54	GPIO54, GMAC_PHY_RXER_PAD, ADC2_CHANNEL5, ANA_COMP1
86	C5_IO23	GPIO23

87	C5_IO24	GPIO24
88	C5_IO25	GPIO25
89	C5_IO26	GPIO26
90	C5_IO27	GPIO27
91	C5_IO28	GPIO28(ESP32-C5 BOOT Pin)
92	GND	GROUND

3.3.Startup Item Configuration

3.3.1. ESP32-P4 Strapping Pins

The chip allows for configuring the following boot parameters through strapping pins and eFuse bits at power-up or a hardware reset, without microcontroller interaction.

- **Chip Boot Mode**

- Strapping pin: GPIO35, GPIO36, GPIO37 and GPIO38

- **ROM Message Printing**

- Strapping pin: GPIO36

- eFuse bit: EFUSE_UART_PRINT_CONTROL

- **JTAG Signal Source**

- Strapping pin: GPIO34

- eFuse bit: EFUSE_DIS_PAD_JTAG, EFUSE_DIS_USB_JTAG and EFUSE_JTAG_SEL_ENABLE

The default values of all the above eFuse bits are 0, which means that they are not burnt. Given that eFuse is one-time programmable, once an eFuse bit is programmed to 1, it can never be reverted to 0.

The default values of the strapping pins, namely the logic levels, are determined by pins' internal weak pull-up/pull-down resistors at reset if the pins are not connected to any circuit, or connected to an external high-impedance circuit.

Default Configuration of Strapping Pin

Strapping Pin	Default Configuration	Value
---------------	-----------------------	-------

GPIO34	Floating	-
GPIO35	Weak pull-up	1
GPIO36	Floating	-
GPIO37	Floating	-
GPIO38	Floating	-

To change the bit values, the strapping pins should be connected to external pull-down/pull-up resistors. If the ESP32-P4 is used as a device by a host MCU, the strapping pin voltage levels can also be controlled by the host MCU.

All strapping pins have latches. At system reset, the latches sample the bit values of their respective strapping pins and store them until the chip is powered down or shut down. The states of latches cannot be changed in any other way. It makes the strapping pin values available during the entire chip operation, and the pins are freed up to be used as regular IO pins after reset.

3.3.2. ESP32-P4 Chip Boot Mode Control

GPIO35 ~ GPIO38 control the boot mode after the reset is released.

Boot Mode	GPIO35	GPIO36	GPIO37	GPIO38
SPI Boot*	1*	Any value	Any value	Any value
Joint Download Boot	0	1	Any value	Any value

*marks the default value and configuration.

Joint Download Boot mode supports the following download methods:

- USB Download Boot:
 - USB-Serial-JTAG Download Boot
 - USB 2.0 OTG Download Boot
- UART Download Boot
- SPI Slave Download Boot

3.3.3. ESP32-P4 ROM Messages Printing Control

During the boot process, the messages by the ROM code can be printed to:

- **(Default) UART0 and USB Serial/JTAG controller**

- USB Serial/JTAG controller
- UART0

EFUSE_UART_PRINT_CONTROL and GPIO36 control ROM messages printing to UART0 as shown in Table.

UART0 Code Printing	EFUSE_UART_PRINT_CONTROL	GPIO36
Enabled*	0*	Ignored
	1	0
	2	1
Disabled	1	1
	2	0
	3	Ignored

*marks the default value and configuration.

EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT controls the printing to USB Serial/JTAG controller as shown in Table.

USB Serial/JTAG ROM Code Printing	EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT
Enabled*	0*
Disabled	1

*marks the default value and configuration.

3.3.4. ESP32-C5 Strapping Pins

The chip allows for configuring the following boot parameters through strapping pins and eFuse bits at power-up or a hardware reset, without microcontroller interaction.

- **Chip Boot Mode**
 - Strapping pin: GPIO26, GPIO27 and GPIO28
- **ROM Message Printing**
 - Strapping pin: GPIO27
 - eFuse bit:

EFUSE_UART_PRINT_CONTROL and EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT

The default values of the strapping pins, namely the logic levels, are determined by pins' internal weak pull-up/pull-down resistors at reset if the pins are not connected to any circuit, or connected to an external high-impedance circuit.

Default Configuration of Strapping Pin

Strapping Pin	Default Configuration	Value
GPIO26	Floating	-
GPIO27	Weak pull-up	1
GPIO28	Weak pull-up	1

To change the bit values, the strapping pins should be connected to external pull-down/pull-up resistors.

All strapping pins have latches. At system reset, the latches sample the bit values of their respective strapping pins and store them until the chip is powered down or shut down. The states of latches cannot be changed in any other way. It makes the strapping pin values available during the entire chip operation, and the pins are freed up to be used as regular IO pins after reset.

3.3.5. ESP32-C5 Chip Boot Mode Control

GPIO27 and GPIO28 control the boot mode after the reset is released.

Boot Mode	GPIO26	GPIO27	GPIO28
SPI Boot*	Any value	Any value	1*
Joint Download Boot 0	Any value	1	0
Joint Download Boot 1	0	0	0

*marks the default value and configuration.

Joint Download Boot mode 0 supports the following download methods:

- USB-Serial-JTAG Download Boot
- UART Download Boot

Joint Download Boot mode 1 supports the following download methods:

- UART Download Boot
- SDIO Download Boot

In SPI Boot mode, the ROM bootloader loads and executes the program from SPI flash to boot the system.

In Joint Download Boot 0 mode, users can download binary files into flash using UART0 or USB interfaces. It is also possible to download binary files into SRAM and execute it from SRAM.

In Joint Download Boot 1 mode, users can download binary files into flash using UART0 or SDIO interfaces. It is also possible to download binary files into SRAM and execute it from SRAM.

3.3.6. ESP32-C5 ROM Messages Printing Control

During the boot process, the messages by the ROM code can be printed to:

- **(Default) UART0 and USB Serial/JTAG controller**
- USB Serial/JTAG controller
- UART0

EFUSE_UART_PRINT_CONTROL and GPIO27 control ROM messages printing to **UART0** as shown in Table.

UART0 Code Printing	Register ²	eFuse ³	GPIO27
ROM messages are always printed to UART0 during boot*	0*	0 (0b00) *	X ⁴
Print is enabled during boot		1 (0b01)	0
Print is disabled during boot			1
Print is disabled during boot		2 (0b10)	0
Print is enabled during boot			1
Print is disabled during boot		3 (0b11)	x
Print is disabled during boot	1	x	x

*marks the default value and configuration.

² Register: LP_AON_STORE4_REG[0]

³ eFuse: EFUSE_UART_PRINT_CONTROL

⁴x: x indicates that the value has no effect on the result and can be ignored

EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT controls the printing to USB Serial/JTAG controller as shown in Table.

USB Serial/JTAG ROM Code Printing	EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT
Enabled*	0*
Disabled	1
	Ignored

*marks the default value and configuration

3.4.Other Pin Descriptions

The GPIO12 and GPIO13 pins of the ESP32-P4 chip on the core board are connected to the GPIO0 and CHIP_PU pins of the ESP32-C5 chip on the core board (see table below). The ESP32-P4 can wake up the ESP32-C5 via GPIO12. and the ESP32-P4 can reset the ESP32-C5 via GPIO13.

ESP32-P4	ESP32-C5
GPIO12	GPIO0
GPIO13	CHIP_PU

The GPIO14, GPIO15, GPIO16, GPIO17, GPIO18, and GPIO19 pins of the ESP32-P4 chip on the core board are connected to the GPIO7, GPIO8, GPIO9, GPIO10, GPIO13, and GPIO14 pins on the ESP32-C5 chip within the core board (see table below). Communication between the two chips occurs via the SDIO interface.

ESP32-P4	ESP32-C5
GPIO14	GPIO8
GPIO15	GPIO7
GPIO16	GPIO14
GPIO17	GPIO13
GPIO18	GPIO9
GPIO19	GPIO10

4. Electrical Characteristics

4.1. Absolute Maximum Limit Value

Exceeding the absolute maximum ratings may result in permanent damage to the device. This is an emphasized rating only and does not address the functional operation of the device under these or other conditions beyond those indicated in these specifications. Prolonged exposure to absolute maximum rating conditions may affect module reliability.

4.2. Power Consumption Characteristics

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4.3. Recommended Working Conditions

Parameter	Description	Min	Typ	Max	Unit
VCC	Power pin voltage	4.8	5	5.5	V
I _{VCC}	Supply current from external power supply	-	1.5	-	A
T _A	Operating Temperature	-40	-	85	°C

5. WT01P4C5-S1 Schematic

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Figure 8: WT01P4C5-S1 Schematic

6. WT01P4C5-S1 Dimensions

The following figure shows the top view and front view of WT01P4C5-S1 with a tolerance of ± 0.2 mm.

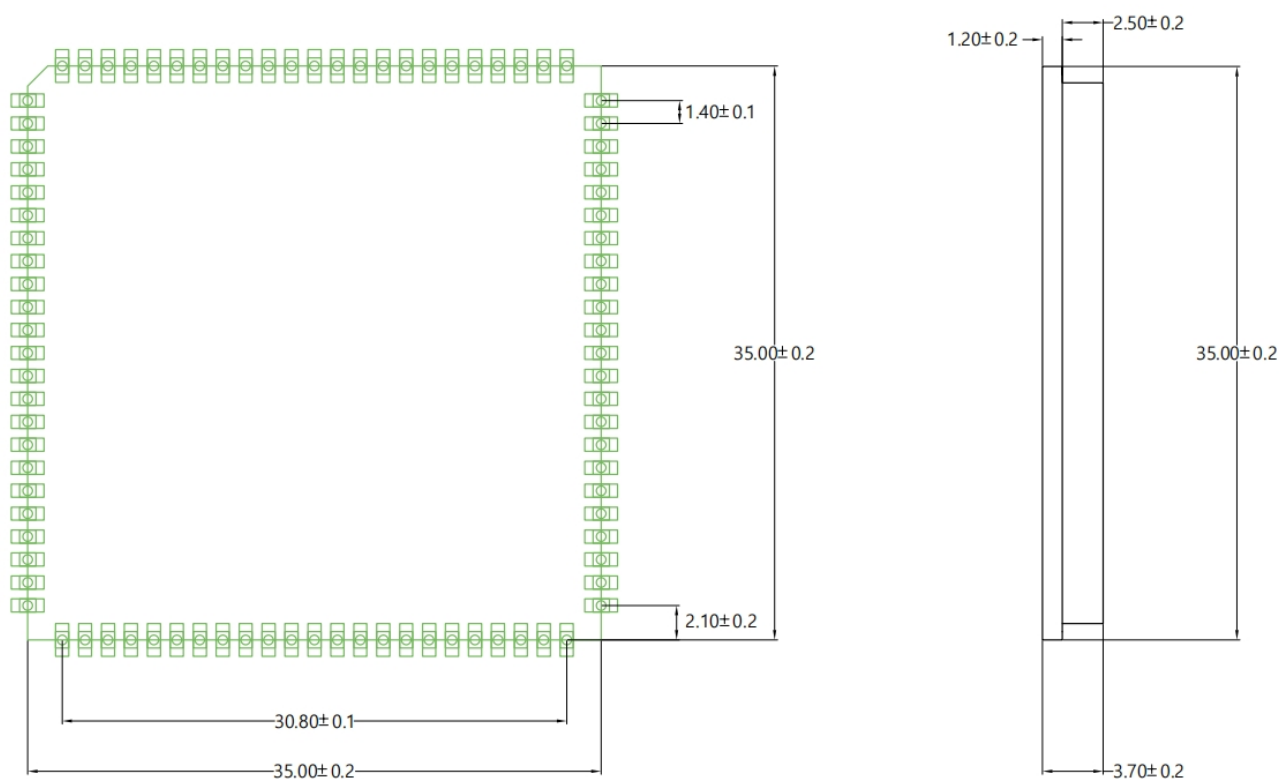


Figure 9: Dimension figure of WT01P4C5-S1

7. Storage Condition

Prerequisite	Parameters
Storage condition	Non-condensing atmosphere < 40°C /90 %RH in sealed MBBs
Conditions of use	168 hours at 25 ± 5°C , 60 % RH.
Moisture sensitivity	3 levels

8. Reflow Soldering Curve

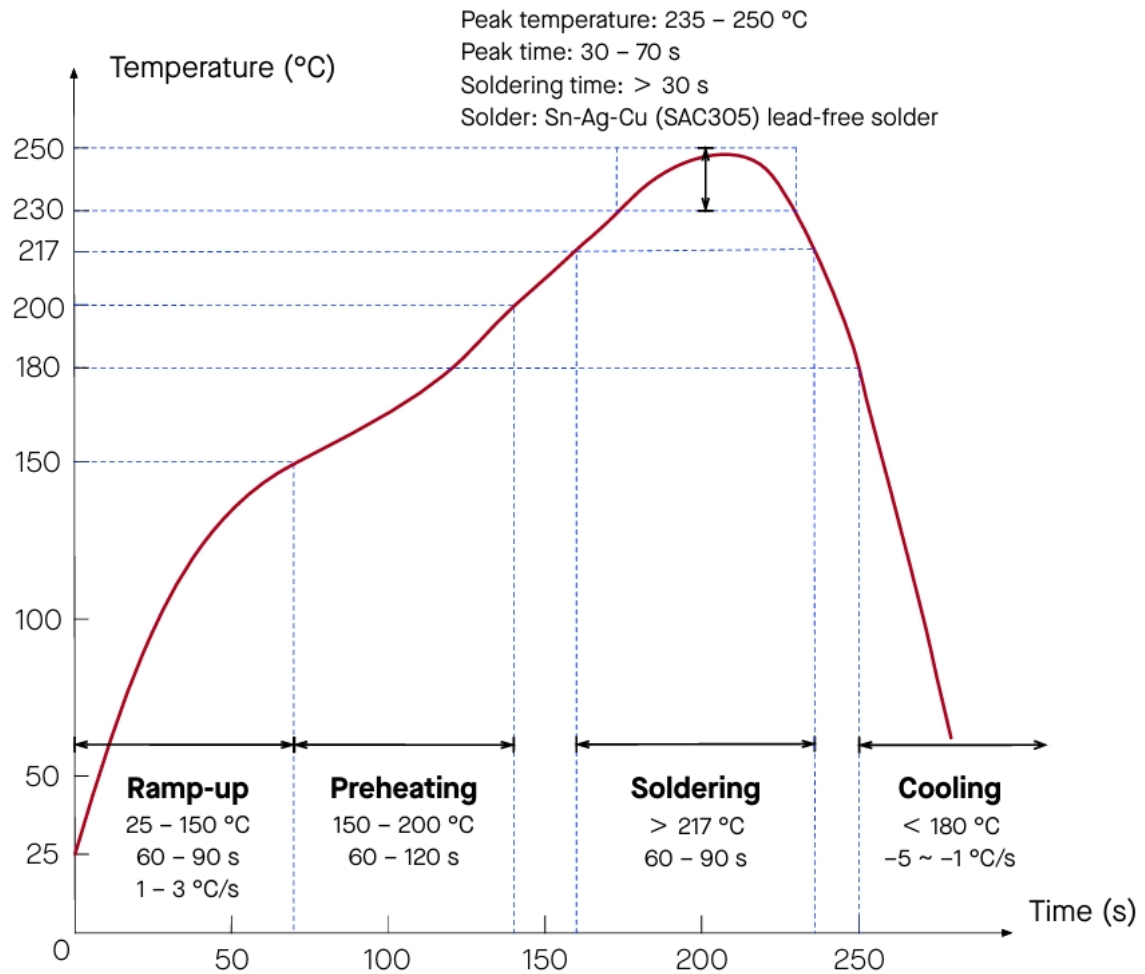


Figure 10: Reflow Soldering Temperature Curve

9. Contact Us

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